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EEL 3135 – Classwork #08

Milestone:

- Complete any 2 questions to receive credit.

Question #1: Compute the following by hand:

$$y[n] = (2\delta[n+1] - \delta[n-2] + 3\delta[n-4]) * (\delta[n-1] + 2\delta[n-3])$$

(a) Write out all shifted and weighted impulse terms explicitly.

$$2\delta[n+1] * \delta[n-1] = 2\delta[n]$$

$$2\delta[n+1] * 2\delta[n-3] = 4\delta[n-2]$$

$$-\delta[n-2] * \delta[n-1] = -\delta[n-3]$$

$$-\delta[n-2] * 2\delta[n-3] = -2\delta[n-5]$$

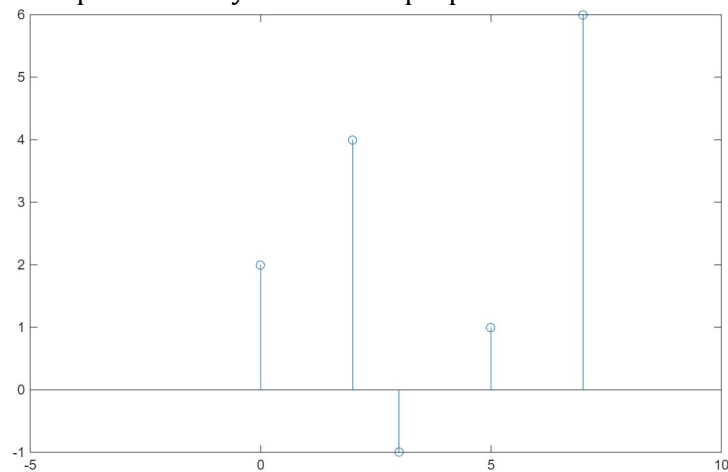
$$3\delta[n-4] * \delta[n-1] = 3\delta[n-5]$$

$$3\delta[n-4] * 2\delta[n-3] = 6\delta[n-7]$$

(b) Simplify to a single expression for $y[n]$ using discrete convolution properties.

$$= 2\delta[n] + 4\delta[n-2] - \delta[n-3] + \delta[n-5] + 6\delta[n-7]$$

(c) Plot the resulting $y[n]$ from part (2) in MATLAB for $n = -5:10$ using the stem() command. Label all nonzero impulses clearly and include proper axis titles.



Question #2: Two FIR systems are connected in cascade:

$$h_1[n] = \frac{1}{2}(\delta[n] + \delta[n-1]), \quad h_2[n] = \delta[n] - \delta[n-2]$$

with input $x[n] = u[n]$.

(a) Compute the overall impulse response $h_T[n] = h_1[n] * h_2[n]$.

$$\frac{1}{2} \delta[n] - \frac{1}{2} \delta[n-2] + \delta[n-1] - \delta[n-3]$$

(b) Determine the system output $y[n] = x[n] * h_T[n]$.

$$\frac{1}{2} u[n] - \frac{1}{2} u[n-2] + u[n-1] - u[n-3]$$

(c) Is the overall system **causal** and **FIR**? Justify your answer.

It is causal because there are no advanced checks ($n+1$, $n+2$, $n+x$)

It is FIR, because it is only non-zero at 0,1,2,3 and then zero.

Question #3: *Given:* Let $x = [1, -1, 2, 0]$ and $h = [2, 1]$.

- (a) Write MATLAB code that constructs the **Toeplitz matrix** for x and computes $y = T_x * h(:)$.
- (b) Compare the result to `conv(x, h)` and confirm they match.
- (c) Briefly explain why both methods yield identical outputs according to the *commutative property* of convolution.

Question #4: *Given:* A discrete-time signal is given as:

$$x[n] = \sin(0.2\pi n) + 0.5w[n],$$

where $w[n]$ is white Gaussian noise. A length-5 moving average FIR filter is applied:

$$h[n] = \frac{1}{5}(u[n] - u[n - 5])$$

- (a) Write MATLAB code to generate $x[n]$, apply the filter using `conv(x,h,'same')`, and plot both signals.
- (b) Estimate qualitatively how the filter affects the frequency content (i.e., explain why it smooths $x[n]$).
- (c) If the filter length is increased to 20, describe the trade-off between **noise reduction** and **signal delay**.